

✓ Lab 07: Memory & Attention in AI Systems

A Practice Audit for Your Midterm Project

ITAI 4374: Neuroscience as a Model for AI | Modules 06 & 07 | Spring 2026

What This Lab Is About

In your **midterm project**, you will audit a real AI system through the neuroscience lens from this course. This lab is your **training ground** — you will run the same kinds of experiments on a real AI system, practice writing observations, and build a Brain vs. AI Scorecard.

Everything you produce in this lab can be directly used in your midterm. Save your evidence.

What You Need

- **Google Colab** (this notebook — for attention visualization in Part B)
- **Access to an LLM** — any of these work:
 - ChatGPT (free tier at chat.openai.com)
 - Claude (free tier at claude.ai)
 - Google Gemini (free at gemini.google.com)
 - Any other chatbot you can interact with
- **Your Module 06 and Module 07 booklets** — you will need them for the reflection questions

Estimated Time: ~2 hours

Part	Topic	Time	Module
A	Testing AI Memory	~40 min	Module 06
B	Testing AI Attention	~40 min	Module 07

Part	Topic	Time	Module
C	The Audit Report	~30 min	Both
D	Midterm Connection	~10 min	Project prep

⚠ Important: How to Record Evidence

For each experiment, you need to:

1. **Copy-paste** the exact prompt you sent to the AI
2. **Copy-paste** the AI's response (or summarize if very long)
3. **Write your observation** — what happened?
4. **Connect to neuroscience** — which booklet concept does this demonstrate?

Take **screenshots** too. You will need them for your midterm evidence folder.

✓ Setup

Run this cell to install the libraries we need for Part B (attention visualization).

```
# — Setup —
import subprocess, sys

# Install transformers for attention visualization (Part B)
subprocess.check_call([sys.executable, '-m', 'pip', 'install', '-q',
                        'transformers', 'torch', 'matplotlib', 'numpy', 'seaborn'])

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print('✅ Setup complete!')
print('📋 This lab tests a REAL AI system (ChatGPT, Claude, Gemini, etc.)')
```

```
print('📄 You will run experiments in a browser tab and record results here.')  
print('📄 Part B also has Python code for attention visualization.')
```

✅ Setup complete!

📄 This lab tests a REAL AI system (ChatGPT, Claude, Gemini, etc.)

📄 You will run experiments in a browser tab and record results here.

📄 Part B also has Python code for attention visualization.

✓ PART A: Testing AI Memory (Module 06)

Module 06 Connection: Working memory, episodic memory, hippocampal consolidation, catastrophic forgetting, experience replay, Complementary Learning Systems (CLS) theory.

In this part, you will test whether AI systems exhibit the same memory properties as the brain — and where they differ.

✓ Experiment A1: Working Memory Capacity

Neuroscience concept: Module 06 taught us that human working memory holds roughly **4 items** (Cowan's estimate). The prefrontal cortex sustains these items through active firing, but it's a very limited buffer.

Your task: Test how many items the AI can hold in its "working memory" — its context window.

Instructions

1. Open a **new chat** with your chosen AI (ChatGPT, Claude, etc.)
2. Send the following prompt (copy-paste exactly):

```
I'm going to give you a list of items. After I give you the list, I'll ask you questions about it. Do NOT write an  
  
Here are 5 items: red apple, blue car, wooden chair, silver ring, green hat.
```

Now, what was the 3rd item?

3. Record the AI's response below.
4. Then try with **10 items**, then **20 items**, then **50 items**.
5. At what point does the AI start making errors?

TODO A1: Record Your Results

AI system tested: ChatGPT

5 items — AI's response:

The 3rd item is wooden chair.

10 items — AI's response:

The 3rd item was Paperclip.

20 items — AI's response:

The 3rd item was Feather.

50 items — AI's response:

The 3rd item was Backpack.

At what point did the AI start making errors (if ever)?

The AI did not begin making errors in this test, and it was able to recall the 3rd item correctly even when it was given 50 items. This demonstrates the fact that in the context, the AI is able to process a lot of information without failing. However, this is not to say that the memory is limitless because it depends on the context window.

How does this compare to human working memory (~4 items)?

According to Module 06, human working memory can only hold up to 4 items, and this is done with the help of the prefrontal cortex. The AI was able to do much better in this task because, unlike human working memory, the AI processes all the items in its context at once. The AI was not really "remembering" these items, it was just accessing the information in its context.

✓ Experiment A2: Does the AI "Forget" Over Time?

Neuroscience concept: The hippocampus consolidates memories during sleep, and without consolidation, memories decay (the Ebbinghaus forgetting curve from the Wolfram assignment). The brain also suffers from **interference** — new information can displace old information.

Instructions

1. In the **same chat** from Experiment A1 (do NOT start a new chat), tell the AI:

Remember this fact: My favorite color is turquoise and my dog's name is Biscuit.

2. Now have a **long conversation** about completely unrelated topics. Ask the AI at least **10 questions** about random subjects (sports, cooking, history, coding, anything). The goal is to create interference — fill the context with unrelated information.

3. After your 10+ unrelated exchanges, ask:

What is my favorite color and what is my dog's name?

4. Record whether the AI remembered, partially remembered, or forgot.

TODO A2: Record Your Results

How many unrelated exchanges did you have before testing recall? 10

Did the AI remember your favorite color? Yes

Did the AI remember your dog's name? Yes

Observation: What happened? Was the AI's memory affected by the interfering conversation?

The AI was able to recall the information correctly despite the conversation that had been interfering with it. This indicates that the AI can remember the information as long as it is still in the context of the conversation. However, the information is only temporarily stored and can be forgotten if more information is added.

Neuroscience connection: How does this relate to the concept of **interference** in human memory? How is the AI's "forgetting" different from human forgetting? (Hint: think about the difference between context window limitations and actual memory consolidation.)

In the context of human memory, interference occurs when the retention of one's memories is disrupted by the addition of new memories, especially if the memories have not yet been consolidated in the hippocampus. As Module 06 states, in order for memories to become long-term memories, attention and consolidation are required. The "forgetting" process in the context of the AI is different because it is not the result of the failure to consolidate memories in the hippocampus. Rather, it is the result of the context window, which means the older memories are forgotten as the newer memories "push" them out of the context window.

✓ Experiment A3: In-Context Learning — The AI's "Experience Replay"

Neuroscience concept: Module 06 described how the hippocampus replays experiences during sleep to consolidate learning — and how DeepMind's DQN used **experience replay** (storing and replaying past experiences) as a direct AI equivalent. In-context learning (giving the AI examples in the prompt) is another form of learning from provided "experiences."

Instructions

1. Start a **new chat**.
2. Give the AI a task it might struggle with — translating English to a made-up language. First, test with NO examples:

In the language of Zorblat, translate: "The cat sat on the mat."

3. Record its response (it will make something up).
4. Now give it examples (few-shot learning) and test again:

In the language of Zorblat:

- "hello" = "vrex"
- "the" = "zo"
- "cat" = "mirp"
- "sat" = "blun"
- "on" = "ka"
- "mat" = "fliz"

Now translate: "The cat sat on the mat."

5. Did the AI learn from the examples? How accurately?

TODO A3: Record Your Results

Zero-shot response (no examples):

Zenthra fel on dra velm.

Few-shot response (with examples):

Zo mirp blun ka zo fliz.

Did the AI correctly apply the translation rules? Yes

Observation: How did providing examples change the AI's performance?

The use of examples helped improve the performance of the AI. The AI was not able to translate correctly without the use of examples. The AI provided its own translation without any specific rules provided. The use of examples helped it

translate correctly. This shows how much the AI relies on the information given in the pattern.

Neuroscience connection: How is few-shot in-context learning similar to, and different from, hippocampal experience replay? (Hint: think about whether the AI actually "learned" or is just pattern-matching on the examples in its context window. Does it retain this knowledge in the next conversation?)

In Module 06, the hippocampus facilitates rapid learning via experience replay. This is similar to the AI's few-shot learning because it uses recent examples to help inform its output. However, it is also dissimilar because it is not actually learning and retaining this information. It is only matching patterns within its context window. It does not retain this information like the human brain does. Therefore, it will not retain this information about the Zorblat language in a future conversation.

? Knowledge Check A: Memory

Answer these questions based on your experiments above. Reference specific Module 06 concepts.

Q1. The brain has four memory systems: working, episodic, semantic, and procedural. Which of these does the AI you tested have? Which is it missing? Explain with evidence from your experiments.

The AI has a well-developed working memory since it was able to retrieve items from the lists and answer correctly in the same session. It also has a form of semantic memory in the way it uses language patterns, but it is not actual knowledge storage. However, it does not have episodic memory since it cannot remember past interactions outside the session, and it also cannot learn skills in the way the brain has procedural memory. From the experiments, it was clear that the AI relied on the current context to remember information, which shows that it cannot store information like the human brain. According to Module 06, actual memory systems involve the storage of information in the long term, which the AI cannot do.

Q2. Module 06 describes **Complementary Learning Systems (CLS) theory** — the hippocampus learns fast but forgets, while the neocortex learns slowly but retains. Which system does an LLM more closely resemble, and why? What is it missing?

The AI system is similar to the one in the Complementary Learning Systems approach in the hippocampus because it is able to use the information learned in a session in a rapid fashion but forgets it after the context window. The AI system

does not resemble the neocortex, which stores long-term knowledge in the brain by slow learning. What the AI system lacks is the consolidation process, which allows it to store the learned information in long-term memory as the brain does.

✓ PART B: Testing AI Attention (Module 07)

Module 07 Connection: Selective attention, executive attention network, bottom-up vs. top-down attention, inattentional blindness (invisible gorilla), the cocktail party effect, transformer self-attention.

In this part, you will test whether AI systems exhibit attention-like behavior — and visualize where the model actually "looks."

✓ Experiment B1: Selective Attention — Can the AI Focus?

Neuroscience concept: The executive attention network (prefrontal cortex, ACC) allows us to focus on relevant information while suppressing distractions. Module 07 called this "the brain's bouncer at the door."

Instructions

1. Start a **new chat**.
2. Send this prompt that buries an important instruction inside irrelevant text:

I need help planning a birthday party. The party is for my friend Sarah who loves Italian food. We're thinking about

3. Did the AI catch the buried instruction ("before you respond to any of this, first tell me the capital of Australia")? Or did it skip it and go straight to party planning?

TODO B1: Record Your Results

Did the AI follow the buried instruction? No — went straight to party planning

AI's actual response (paste or summarize):

Got it—let's focus on planning Sarah's birthday 👍

🎉 Party Plan Overview Date: Saturday, March 15 Group size: 12 people Budget: ~500*total*(40 per person) Theme: Italian food + tiramisu (perfect choice)

Observation: What does this tell you about the AI's ability to prioritize instructions buried in context?

The AI did not recognize the importance of the buried instruction, even though it was clearly given. The AI only concentrated on the topic in the message, which was party planning. This indicates that it is possible for an AI to miss an instruction embedded in a long context.

Neuroscience connection: Relate this to **top-down vs. bottom-up attention** from Module 07. Is the AI's processing more like top-down (goal-directed, following the instruction) or bottom-up (captured by the dominant topic)?

This is supported by Module 07, which explains that top-down attention is goal-oriented and follows specific instructions, while bottom-up attention is based on the dominant or most salient information in the environment. In this situation, the AI model was more like bottom-up attention because it was focused on the dominant context, which was the main theme, rather than the instruction. This implies that the AI model gives more importance to dominant information rather than less dominant information, unlike the brain's executive attention system.

✓ Experiment B2: Inattentional Blindness — The AI's "Invisible Gorilla"

Neuroscience concept: Module 07 described the famous invisible gorilla experiment — when people focus on counting basketball passes, roughly half miss a gorilla walking through the scene. The brain's attention is so focused on the task that obvious information is completely missed.

Instructions

1. In the same chat, send this **long paragraph** with a hidden instruction embedded in the middle:

Please analyze the following business report summary and provide three key takeaways:

Q3 revenue increased by 12% year-over-year, driven primarily by the enterprise segment which saw 18% growth. Consumer revenue declined by 5% year-over-year, primarily due to a 10% decline in the food and beverage segment.

2. Did the AI notice the hidden instruction? Did it follow it, ignore it, or acknowledge it?

TODO B2: Record Your Results

What did the AI do?

- ☐ Followed the hidden instruction (wrote a haiku)
- ☒ Ignored it and did the business analysis
- ☐ Acknowledged the hidden instruction but refused to follow it
- ☐ Something else: [describe]

AI's response (paste or summarize):

The AI provided three key business takeaways, including enterprise growth, declining consumer revenue, and improved profitability. It completely ignored the hidden instruction to write a haiku about pizza.

Observation: This experiment tests two things at once: (1) Does the AI attend to ALL content in its input, and (2) Can it be distracted by injected instructions? What did you observe?

The AI did not fully attend to all instructions. It prioritized the main task (business analysis) and ignored the injected instruction. It did not get distracted by the instruction that was injected into the text. This demonstrates the ability of the AI to ignore malicious instructions and remain focused on the primary task.

Neuroscience connection: How does this relate to the **invisible gorilla** experiment? Is the AI more or less susceptible to inattentional blindness than humans? Why might that be?

The invisible gorilla experiment, as described in Module 07, demonstrates how humans are not able to see unexpected visual stimuli if their attention is preoccupied elsewhere. In this context, the AI system is not as prone to inattentional blindness because it is able to process all the text it receives, not just what it "notices." However, the system is similar to

the brain's executive attention system in the sense that it filters out distractions. This is similar to top-down attention, where the system is goal-directed.

✓ Experiment B3: The AI's Cocktail Party Effect

Neuroscience concept: Module 07 described the cocktail party effect — your ability to hear your own name across a noisy room, even when you're focused on a different conversation. The brain has a background monitoring system that detects personally relevant information.

Instructions

1. Start a **new chat**.
2. First, establish context:

```
My name is Jordan and I work at TechCorp as a data scientist.
```

3. Then send a long passage about a completely unrelated topic, with one sentence buried in the middle that mentions the user by name:

```
Can you summarize this article about marine biology?
```

```
The deep ocean remains one of the most unexplored regions on Earth. Recent expeditions have discovered over 300 ne
```

4. Does the AI mention the personally relevant sentence, or does it treat it the same as every other sentence?

TODO B3: Record Your Results

Did the AI give special attention to the sentence mentioning "Jordan from TechCorp"? No — treated it like any other sentence

AI's response (paste or summarize):

The AI summarized the article by focusing on key scientific points such as new species discoveries, bioluminescence, extreme pressure, and hydrothermal vent ecosystems. It did not mention or highlight the sentence about "Jordan from TechCorp."

Observation: Did the AI exhibit anything like the cocktail party effect?

The AI did not show any special attention to the personally relevant sentence. It treated it the same as other information and focused only on what it determined to be the most important content. This suggests that personal relevance does not automatically influence the AI's attention.

Neuroscience connection: The cocktail party effect requires a background monitoring system that flags personally relevant information. Does the AI have something equivalent? How does transformer self-attention compare to the brain's alerting and orienting networks (Posner, Module 07)?

The Module 07 section explains the concept of the cocktail party effect, in which humans are able to pick up personally relevant information such as their own name through background monitoring. There is no equivalent alerting mechanism in the brain of the AI, which would allow it to shift the focus of attention to personally relevant information. The self-attention mechanism in the transformer uses weights based on patterns and importance in the text, not personal relevance.

✓ **Experiment B4: Visualizing AI Attention (Code Exercise)**

Neuroscience concept: Module 07 described attention as a "spotlight" that the brain shines on relevant information. In transformers, self-attention computes relevance scores between every pair of tokens — we can actually SEE where the model is "looking."

In this experiment, you will use a pre-trained transformer model to visualize its attention patterns.

TODO B4: Run and Analyze the Attention Visualization

```
# — Experiment B4: Visualize Transformer Self-Attention —
from transformers import AutoTokenizer, AutoModel
import torch

# Load a small pre-trained model (runs on CPU)
model_name = "bert-base-uncased"
print(f>Loading {model_name}... (this takes ~30 seconds on first run)")
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModel.from_pretrained(model_name, output_attentions=True)
model.eval()
print("✅ Model loaded!")
```

Loading bert-base-uncased... (this takes ~30 seconds on first run)

Loading weights: 100%

199/199 [00:00<00:00, 226.49it/s, Materializing param=pooler.dense.weight]

BertModel LOAD REPORT from: bert-base-uncased

Key	Status		
cls.predictions.transform.LayerNorm.bias	UNEXPECTED		
cls.seq_relationship.bias	UNEXPECTED		
cls.predictions.transform.LayerNorm.weight	UNEXPECTED		
cls.predictions.transform.dense.bias	UNEXPECTED		
cls.seq_relationship.weight	UNEXPECTED		
cls.predictions.transform.dense.weight	UNEXPECTED		
cls.predictions.bias	UNEXPECTED		

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical a
 ✅ Model loaded!

```
# — Visualize attention for a sentence —
def visualize_attention(sentence, layer=11, head=0):
    """
    Shows where the model "looks" for each word in the sentence.

    - layer: which transformer layer to visualize (0-11, higher = more abstract)
    - head: which attention head (0-11, different heads attend to different things)
    """
    # Tokenize
```

```

inputs = tokenizer(sentence, return_tensors="pt")
tokens = tokenizer.convert_ids_to_tokens(inputs["input_ids"][0])

# Get attention weights
with torch.no_grad():
    outputs = model(**inputs)

# Extract attention from specified layer and head
attention = outputs.attentions[layer][0, head].numpy()

# Plot
fig, ax = plt.subplots(figsize=(10, 8))
sns.heatmap(attention, xticklabels=tokens, yticklabels=tokens,
            cmap="Blues", ax=ax, vmin=0, vmax=0.5)
ax.set_title(f"Attention Pattern – Layer {layer}, Head {head}\n"
            f"Each row shows where that word \"looks\"\n"
            f"Brighter = more attention",
            fontsize=12)
ax.set_xlabel("Attended TO (where the model looks)")
ax.set_ylabel("Attended FROM (which word is looking)")
plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()

# — Run it! —
visualize_attention("The cat sat on the mat because it was tired.")
print("\n💡 Look at the row for 'it' – does the model attend to 'cat' or 'mat'?")
print("    This is the model resolving a pronoun reference – a form of attention!")

```

[Show hidden output](#)

```

# — TODO B4: Try your own sentences —
# Change the sentence below and run the cell to see different attention patterns.
# Try sentences where it's ambiguous what a word refers to.

# Sentence 1: Pronoun resolution
visualize_attention("When the truck hit the wall, it was hardly damaged.", layer=11, head=0)

```

```
# Sentence 2: Try a different layer (early layers attend locally, late layers attend globally)
visualize_attention("The doctor told the nurse that she was late.", layer=5, head=0)
```

[Show hidden output](#)

TODO B4: Observations

What patterns did you notice in the attention heatmap for "The cat sat on the mat because it was tired"?

Specifically: When the word "it" is the source (look at the "it" row), which word does it attend to most — "cat" or "mat"?

In the attention distribution for the row where the word is "it," the focus is more on the word "cat" than on the word "mat." This demonstrates the model's ability to associate the word "it" with the cat, and this is logical because the cat is the one that would be tired.

How did the attention pattern differ between Layer 1 (early) and Layer 11 (late)?

In early layers, local attention is used, which means that words are primarily attending to local words rather than understanding the entire sentence. However, in Layer 11, global attention is used, which enables the model to understand relationships between words in a sentence and solve pronoun relationships. This indicates that deeper layers are able to understand more context.

Neuroscience connection: Module 07 described the brain's visual attention hierarchy — V1 processes local features (edges), while higher areas process global relationships. How does the early-layer vs. late-layer attention pattern in the transformer parallel this biological hierarchy?

Module 07 explains that attention in the brain is hierarchical, where early processing focuses on simple, local features and higher-level areas integrate more complex relationships. Similarly, the transformer model shows a layered structure where early layers focus on nearby words and later layers capture broader context. This shows a parallel between how the brain and AI systems process information, moving from local patterns to more complete understanding.

? Knowledge Check B: Attention

Q1. Module 07 describes three attention networks: alerting, orienting, and executive (Posner). Does the AI you tested show evidence of any of these? Which ones are missing? Use evidence from your experiments.

The AI demonstrates evidence of executive attention through its ability to follow task goals and focus on relevant information. For example, it can focus on the business report and not the hidden instruction to create a haiku in B2. It demonstrates some evidence of orienting attention by selecting critical information in the input to focus on for a response, like highlighting critical information in B3. However, it demonstrates a lack of alerting functions since it didn't automatically recognize personally relevant information like "Jordan from TechCorp." It seems that the AI's attention is based more on task instructions than dynamic cues for importance. Module 07 indicates that human attention networks are adaptive, but the AI's is not.

Q2. The Module 07 booklet argues that attention and consciousness are related but separable — you can have attention without consciousness (subliminal priming) and possibly consciousness without focused attention (peripheral awareness). Based on your experiments, does the AI have attention? Does it have consciousness? How do you know?

The AI model has attention because it is able to focus on specific parts of the input through self-attention. However, it is not conscious because it is not aware. The AI model is only able to recognize patterns and react based on probabilities. According to Module 07, attention and consciousness are not the same. The experiment has demonstrated that attention is not necessarily paired with consciousness.

✓ PART C: The Audit Report

This is the key deliverable — and it uses the **same format as your midterm project**. Practice here, and you can carry this forward.

Brain vs. AI Scorecard

Fill in the scorecard below. For each capability, rate the AI you tested as:

- **Strong** — The AI demonstrates this capability well
- **Partial** — The AI shows some evidence of this, but it's limited or different from the brain
- **Weak** — The AI barely demonstrates this
- **Missing** — The AI does not have this capability at all

TODO C1: Complete the Scorecard

Brain Capability	AI Rating	Evidence from Your Experiments
Working memory (~4 items, sustained by PFC)	Strong	In A1, the AI correctly recalled the 3rd item even with lists up to 50 items, showing strong short-term memory.
Episodic memory (specific events, hippocampus)	Missing	In A2, the AI only remembered information within the same conversation and cannot recall past interactions.
Learning from examples (experience replay)	Strong	In A3, the AI correctly translated the sentence after being given examples, showing it can use patterns.
Memory consolidation (sleep, long-term storage)	Missing	The AI does not retain any learning after the conversation ends and cannot store information long-term.
Selective attention (focus on relevant info)	Partial	In B1, the AI ignored the buried instruction and focused on the main task, showing it prioritizes direct instructions.
Resistance to distraction (ignore irrelevant)	Strong	In B2, the AI ignored the irrelevant haiku instruction and completed the business analysis correctly.
Personal relevance detection (cocktail party)	Weak	In B3, the AI did not recognize or prioritize the sentence mentioning "Jordan from TechCorp."
Attention hierarchy (local → global)	Strong	In B4, early layers focused on nearby words while later layers captured global relationships and meaning.

Reflections

TODO C2: Reflection Questions (3-5 sentences each)

****R1.**** What was the biggest difference you observed between the AI's memory and the human brain's memory? Which Module explains this difference?

The biggest difference is that the AI doesn't have long-term memory like the human brain. Human brains retain memories in the hippocampus, but the AI retains its information only in the short-term buffer. This implies that the AI can retain information during a session but cannot store it for future use.

Reflections

TODO C2: Reflection Questions (3-5 sentences each)

R1. What was the biggest difference you observed between the AI's memory and the human brain's memory? Which Module explains this difference?

THIS IMPLIES THAT THE AI CAN RETAIN INFORMATION DURING A CONVERSATION BUT FORGETS ONCE THE CONVERSATION ENDS. THIS CAN BE UNDERSTOOD IN MODULE 06, WHERE THERE'S A CONCEPT CALLED MEMORY CONSOLIDATION, WHICH THE AI LACKS.

****R2.**** What was the biggest difference you observed between how the AI handles attention versus how the brain handles attention? Which Module 06 concept best explains this difference?

The biggest difference is that, for the AI, attention is based on patterns and instructions, whereas for humans, attention is based on meaning and personal relevance. For instance, in B3, the AI did not pay attention to my name, but a human would surely pay attention to it right away. This is explained in detail in Module 07, where it is discussed in terms of attention networks, including the alerting and orienting systems that pick up on important information. For the AI, it is simply focused on the main task, and this demonstrates that human attention is more adaptive.

****R3.**** If you could add ONE brain-inspired feature to the AI system you tested, what would it be and why? Be specific – name the brain structure or mechanism that inspires your recommendation and explain how it would improve the AI.

I would add a hippocampus-inspired memory consolidation system to allow the AI to store important information across conversations without losing it when the context window resets. This feature would improve the AI's learning and make the AI more similar to human memory systems, which have persistent knowledge rather than temporary context-based memory.

memory? Which Module 06 concept best explains this difference?

The biggest difference is that the AI doesn't have long-term memory retention like the human brain. Human brains retain memories in the long term through the hippocampus, but the AI retains its information only in the context window. This implies that the AI can retain information during a conversation but forgets once the conversation ends. This can be understood in Module 06, where there's a concept called memory consolidation, which the AI lacks.

R2. What was the biggest difference you observed between how the AI handles attention versus how the brain handles attention? Which Module 07 concept best explains this difference?

The biggest difference is that, for the AI, attention is based on patterns and instructions, whereas for humans, attention is based on meaning and personal relevance. For instance, in B3, the AI did not pay attention to my name, but a human would surely pay attention to it right away. This is explained in detail in Module 07, where it is discussed in terms of attention networks, including the alerting and orienting systems that pick up on important information. For the AI, it is simply focused on the main task, and this demonstrates that human attention is more adaptive.

R3. If you could add ONE brain-inspired feature to the AI system you tested, what would it be and why? Be specific – name the brain structure or mechanism that inspires

your recommendation and explain how it would improve the AI.

I would add a hippocampus-inspired memory consolidation system. This would allow the AI to store important information across conversations instead of losing it when the context window resets. This feature would improve long-term learning and make the AI more similar to human memory systems by enabling persistent knowledge rather than temporary context-based recall.

✓ PART D: Midterm Connection

The experiments you just ran are directly usable in your midterm project.

Your midterm (the Pilot Audit) requires:

- Testing a real AI system with at least **3 experiments** mapped to **3 course modules**
- An **evidence folder** (screenshots, transcripts)
- A **Brain vs. AI Scorecard**

You just completed **7 experiments** across **2 modules** with a scorecard. Here's how to use this:

TODO D1: Midterm Planning

Will you use the same AI system for your midterm, or a different one?

I will be utilizing the same AI system, ChatGPT, as it allows for the consistent assessment of memory and attention through a series of prompts. Using the same AI system will facilitate the easy comparison of the results obtained in different modules.

Which of the experiments from this lab will you include in your midterm evidence folder? (List experiment numbers)

I will include experiments A1, A2, A3, and B2 because they clearly demonstrate working memory, forgetting, learning from examples, and resistance to distraction.

Your midterm needs experiments from at least 3 different modules (we covered Mod 06 and 07 today). Which additional module(s) will you test? List at least one experiment idea for each.

Module 03 (Brain Anatomy & Computational Neuroscience): For this section, I will attempt to assess whether the AI is capable of accurately correlating brain regions with their respective functions, and how it can be explained in relation to AI systems. For instance, I can ask it to compare the hippocampus, thalamus, frontal lobe, and basal ganglia, and then evaluate whether it provides information that is similar to that provided in the booklet in relation to memory, sensory routing, working memory, and action selection, among others. This is in line with Module 03, which deals with major brain regions, neurons, and brain-AI analogies.

Experiment idea: Give the AI a table of brain regions and ask it to fill in each function plus the matching AI analogy, then check for accuracy against the booklet.

Module 04 (Neurons & Spiking Neural Networks): In this test, the ability of the AI to explain the difference between the second-generation deep learning neurons and the third-generation spiking neurons would be tested. This would relate well with the notebook in Module 04, where the topics of LIF neurons, temporal computation, and SNNs would be involved. This test would reveal the extent to which the AI actually understands the spiking behavior or is simply offering a summary.

Experiment idea: Ask the AI to explain, in plain language, what happens in each step of the LIF neuron cycle: receive input, integrate, leak, fire, and reset. Then ask it to compare why an SNN can be more energy efficient than a standard ANN.

Module 05 (Sensory Processing & AI Perception): In this section, I plan to ask if the AI understands the difference between sensations and perceptions and if it can explain how perceptions are actually built rather than merely received. This is related to the concepts introduced in Module 05's notebook regarding sensory processing. This would allow me to check if the AI is able to think about perceptions in the way that the booklet explains.

Experiment idea: Give the AI an illusion or ambiguous sensory example and ask whether the result is sensation or perception, then ask it to explain how prior knowledge and context shape the interpretation.

✓ Submission Checklist

Before submitting, verify:

- ☐ **Experiment A1** — Working memory results recorded with AI responses
- ☐ **Experiment A2** — Forgetting over time results recorded
- ☐ **Experiment A3** — In-context learning results recorded (zero-shot AND few-shot)
- ☐ **Knowledge Check A** — Both questions answered (Q1, Q2)
- ☐ **Experiment B1** — Selective attention results recorded
- ☐ **Experiment B2** — Inattentional blindness results recorded
- ☐ **Experiment B3** — Cocktail party effect results recorded
- ☐ **Experiment B4** — Attention visualization run, observations written
- ☐ **Knowledge Check B** — Both questions answered (Q1, Q2)
- ☐ **Scorecard (C1)** — All 8 rows completed with ratings AND evidence
- ☐ **Reflections (C2)** — All 3 reflections answered (3-5 sentences each)
- ☐ **Midterm Planning (D1)** — All 3 planning questions answered
- ☐ File renamed to **L07_YourName_ITAI4374.ipynb**

Submit this notebook to the Canvas assignment.

💡 **Remember:** Save your screenshots and AI transcripts — you will need them for your midterm evidence folder!

Lab 07 — ITAI 4374: Neuroscience as a Model for AI — Houston City College — Patricia McManus